

CHAN BASHA BADVEL

AI & Robotics Engineer | Java Developer

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UK Graduate Visa (valid until 2028)

PROFESSIONAL SUMMARY

MSc Artificial Intelligence and Robotics graduate (with Advanced Research Commendation) from the University of Hertfordshire, combined with 15 months of professional Java backend development experience at Infosys. Skilled in building end-to-end systems across the stack — from REST APIs and Spring Boot microservices to autonomous robotics integrating Raspberry Pi, Arduino, computer vision, and AI. Eager to apply both software engineering discipline and applied AI/robotics experience in a UK-based engineering role.

EDUCATION

MSc Artificial Intelligence and Robotics — University of Hertfordshire Feb 2026

Awarded with Advanced Research Commendation

Sep 2023 – Feb 2026

- Relevant modules: Neural Networks and Machine Learning (72), Artificial Life with Robotics (77), Team Research and Development Project (78), Theory and Practice of AI (51), Responsible Technology (70), Research Methods, Advanced Research Topics in Computer Science.

Bachelor of Science (BSc)

Nov 2020

S.V. University, Tirupati, Andhra Pradesh, India

Jun 2017 – Nov 2020

- CGPA: 7.7 / 10. Subjects: Mathematics, Physics, Computer Science.

TECHNICAL SKILLS

Programming Languages: Java, Python, C++ (Arduino/embedded), SQL, HTML, CSS.

AI / Machine Learning: Machine Learning, Deep Learning, Neural Networks, Linear Regression, Unsupervised Clustering, Computer Vision, Medical Image Analysis, scikit-learn, pandas, NumPy, matplotlib, OpenCV, Vosk (speech-to-text), pyttsx3 (text-to-speech), GPT-4 / LLM API integration.

Backend & Web: Spring Boot, REST APIs, Hibernate, MySQL, Apache Tomcat, WSO2 Identity Server, Apache Syncope.

Robotics & Hardware: Raspberry Pi 4, Arduino Mega, GPIO, PWM motor control (L298N), ultrasonic sensors (HC-SR04), CSI camera modules, SPI / I2C / I2S / USB serial protocols, LiPo power management, ALSA audio.

Tools & Platforms: Git, Docker, Postman, SoapUI, Eclipse, VS Code, Linux (Raspberry Pi OS / Debian), MS Office.

Methodologies & Research: Agile / Scrum, software development lifecycle, REST API design, unit testing, literature review (PICO framework), human-robot interaction (HRI) research.

PROJECTS

ARIA Bot — AI-Assisted 4WD Autonomous Robot Car

2025 – Present

Independent Project — Raspberry Pi 4, Arduino Mega, Python, C++, OpenCV

- Designing and building an autonomous 4WD robot car from individual components, owning the full stack from hardware wiring and firmware to AI software integration.
- Engineered the 11.1V 3S LiPo power distribution with dual buck converters, wired dual L298N drivers to 4× TT DC gear motors (independent PWM speed/direction), and integrated 3× HC-SR04 ultrasonic sensors in a 180° front arc — managing GPIO across 6 subsystems (SPI/I2C/I2S/CSI/USB serial/PWM) with zero pin conflicts.
- Developed a custom Python ↔ Arduino C++ serial protocol at 115200 baud with a thread-safe Python control library, configured the WM8960 audio HAT via ALSA / device-tree overlays, and built

live H.264 camera streaming over TCP from the Pi Camera Module; AI software stack (Vosk STT → GPT-4 → pyttsx3 TTS, OpenCV target-following) under active development.

Research Methods Project — Medical Image Enhancement

2025

University of Hertfordshire — Literature Review Protocol (Interim Report: 87 / 100)

- Authored a structured literature review protocol on enhancing diabetic retinopathy fundus images using unsupervised clustering techniques to support disease detection.
- Formalised the research question with the PICO framework and critically compared clustering-based enhancement against traditional preprocessing methods (histogram equalisation, CLAHE, contrast stretching).

Specialism Research Report — Human-Robot Interaction

2025

University of Hertfordshire — Advanced Research Topics in CS (graded 70 / 100)

- Produced a critical review of Holthaus & Wachsmuth's holistic encounter model for social robotics, evaluating its non-verbal signal portfolio (gaze, body orientation, gestures, proxemics, spatial prompting) tested on the iCub humanoid receptionist platform.
- Formulated an original open research question on how encounter-level non-verbal behaviours influence user trust, comfort, and re-engagement across repeated interactions, identifying a clear gap in long-term HRI literature.
- Surveyed and synthesised foundational HRI literature spanning proxemics (Hall), f-formations and gaze (Kendon), greetings (Heenan et al.), spatial prompting (Green & Hüttenrauch), and closing behaviour (Rollet & Licoppe).

WORK EXPERIENCE

Java Developer (Systems Engineer)

Feb 2022 – May 2023

Infosys Ltd.

Hyderabad, India

Project: IDAM (Identity and Access Management)

- Explored, analysed, and tested APIs of Apache Syncope and WSO2 Identity Server to ensure seamless integration with existing enterprise systems, reducing access-management errors by ~30%.
- Designed and executed test cases for user provisioning, role mapping, and attribute synchronisation using Postman and SoapUI, cutting user-setup time by ~20%.
- Built backend components in Java, Spring Boot, and REST APIs against a MySQL backing store on Apache Tomcat, contributing to a ~15% improvement in overall system performance, working in an Agile cross-functional team.

Trainee Software Engineer (Internship)

Feb 2022 – Apr 2022

Infosys Ltd.

Hyderabad, India

Project: Internship Management Application

- Developed a Java Spring Boot REST API backend with MySQL persistence and core CRUD modules for an internal training-period web application; completed structured technical training in Java SE, Spring Boot, web services, and Agile delivery.

Part-Time Retail Sales Assistant

Dec 2024 – Present

Nisa Local — Grange Convenience Store, Letchworth, UK

- Worked part-time alongside the MSc programme, serving customers on the till and shop floor, handling cash and card payments, training new colleagues, and opening/closing the shop — demonstrating strong time management, reliability, and customer-service skills.

CERTIFICATIONS & LANGUAGES

Certifications: Infosys Certified Java SE 8, Java SE 11, and Global Agile Developer.

Languages: English (Full Professional), Telugu (Native), Hindi (Full Professional).